Kevin Chen

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Research Interests

Mechanistic Interpretability, Physical World Models, Cross-Modal Generative AI, Radar-Based Human Sensing

Education

- **The Ohio State University** — **Columbus, OH**
  Ph.D. in Computer Science and Engineering — *2022–Dec. 2027 (Expected)*
  - **Advisor:** Dr. Anish Arora
- **National Taiwan University** — **Taipei, Taiwan**
  M.S. in Electrical Engineering — *2015–2017*
- **National Cheng Kung University** — **Tainan, Taiwan**
  B.S. in Photonics — *2011–2015*

Research and Industry Experience

- **The Ohio State University** — **Columbus, OH**
  Graduate Research Assistant — *Aug. 2022–Present*

  Primarily working on physics-grounded mechanistic interpretability for radar generative models (Three first-author publications).
  - Developed a spatiotemporal transformer, MoCap2Radar, which synthesizes micro-Doppler radar signatures from human motion capture data. (HICSS-59)
  - Analyzed whether MoCap2Radar models has learned Doppler radar physics through behavioral and explainability-based tests. (RadarConf 2026)
  - Proposed a physics-grounded interpretability framework for testing whether neural networks realize physical laws. (ICML Workshop 2026)
- **CITI, Academia Sinica** — **Taipei, Taiwan**
  Research Assistant (Mentor: Dr. Ronald Y. Chang) — *2019–2021*

  Developed ML methods and interpretability techniques for WiFi signal-based indoor localization. (Three first-author publications)
- **Automated Verification & Software Testing Lab, NTU** — **Taipei, Taiwan**
  Graduate Research Assistant — *2015–2017*

  Developed methods for automated sandbox deployment in DevOps.
- **Taipei Digital Group** — **Taipei, Taiwan**
  Data Science Intern — *Summer 2016*

  Developed ML methods for content classification to improve search engine optimization.

Workshop Papers

[W1] **Kevin Chen**, Kenneth W. Parker, and Anish Arora, "LAWFUL: Law-Aligned Witness for Faithful Use of Latents," *Mechanistic Interpretability Workshop at ICML 2026* **[In-Person Poster]**.

Conference Publications

[C1] **Kevin Chen**, Kenneth W. Parker, and Anish Arora, "What Physics do Data-Driven MoCap-to-Radar Models Learn?," *IEEE Radar Conference (RadarConf 2026)*.

[C2] **Kevin Chen**, Kenneth W. Parker, and Anish Arora, "MoCap2Radar: A Spatiotemporal Transformer for Synthesizing Micro-Doppler Radar Signatures from Motion Capture," *59th Hawaii International Conference on System Sciences (HICSS-59 2026)*.

[C3] **Kevin Chen** and R. Y. Chang, "A Comparative Study of Deep-Learning-Based Semi-Supervised Device-Free Indoor Localization," *IEEE Global Communications Conference (GLOBECOM 2021)*.

[C4] **Kevin Chen** and R. Y. Chang, "Semi-Supervised Learning with GANs for Device-Free Fingerprinting Indoor Localization," *IEEE Global Communications Conference (GLOBECOM 2020)*.

Journal Publication

[J1] **Kevin Chen**, R. Y. Chang, and S.-J. Liu, "Interpreting Convolutional Neural Networks for Device-Free Wi-Fi Fingerprinting Indoor Localization via Information Visualization," *IEEE Access*, vol. 7, pp. 172156–172166, 2019.

Academic Services

- **Conference Reviewer**
  IEEE International Conference on Communications (2020, 2021)